

Decent Coaching Classes

VIII SB

Academic Conquest - II

Answer Key: Geography - Chapter 4 (Structure of Ocean Floor)

Q.1. Choose the correct option:

(5)

- 1) b) continental shelf
- 2) c) Mariana Trench
- 3) b) 3700
- 4) a) Marine oozes
- 5) c) 8848

Q.2. True / False:

(5)

- 1) False. Correct statement: The continental shelf is the shallowest part of the ocean bed.
- 2) True.
- 3) True.
- 4) True.
- 5) False. Correct statement: Waste materials disposed by man in the oceans are extremely harmful to marine life.

Q.3. Give geographical reasons (any THREE):

(6)

- 1) The continental shelf is a paradise for fishing activity — As the continental shelf is shallow, sunlight reaches its bed, allowing algae and plankton to grow, which serve as food for fish. Hence fish gather here in large numbers, forming rich fishing grounds.
- 2) Some marine islands are actually the peaks of submerged mountains — There are submerged hills and mountains on the ocean bed. When the peaks of some of these submerged mountains rise above sea level, they appear to us as marine islands (e.g. Iceland, Andaman and Nicobar Islands).
- 3) The continental slope is considered to be the boundary of continents — Beyond the continental slope the structure of the sea bed changes into the abyssal plains, which are a part of the deep ocean floor and not of the continent. Hence the lower edge of the continental slope marks where the continent ends.
- 4) The disposal of waste materials in the oceans by man is harmful to the environment — Sewage, solid waste, radioactive material, chemicals and plastics dumped by man are hazardous to the ocean bed and sea water, and extremely harmful to marine life and its ecosystem, even though their proportion is small.

Q.4. Answer in brief (any THREE):

(9)

- 1) Land submerged below oceanic water is called the ocean floor. Its relief is decided on the basis of (i) the depth from the sea surface and (ii) the shape/structure of the submerged land.
- 2) The continental shelf is the land near the coast that is submerged under the sea; it is the shallowest part of the ocean bed with a gentle slope, extending up to about 200 m depth. Its extent varies from place to place. It is important to man because it provides extensive fishing grounds (algae and plankton grow here due to sunlight) and yields natural gas, mineral oil and minerals, e.g. Mumbai High.
- 3) Landforms on the ocean bed that are deep, narrow and steep are called marine deeps or trenches. Generally, shallower ones are called marine deeps, while deeper ones extending over longer distances are called trenches. The Mariana Trench in the Pacific Ocean (about 11,034 m deep) is the deepest trench in the world.
- 4) Mean Sea Level is the average of the highest high tides and the lowest low tides at a place; this average is taken as zero. The elevation or depth of any place is measured with reference to this zero level — heights above it are shown as positive values (e.g. Mount Everest, 8848 m) and depths below it as negative values (e.g. Mariana Trench, -11034 m).

Q.5. Answer in detail (any ONE):

(5)

1) As we move away from the coast, the ocean floor shows the following sequence of relief features:

- Continental Shelf – the shallow, gently sloping submerged coastland near the coast (up to about 200 m depth).
- Continental Slope – beyond the shelf, the sea bed becomes steeper, extending from about 200 m to 3600 m depth; its lower boundary marks the edge of the continent.
- Abyssal Plains – the flat part of the deep sea bed beyond the continental slope, containing submerged hills, plateaus, etc.
- Mid-Oceanic Ridges and Submerged Mountains – long submerged mountain ranges on the ocean bed; peaks that rise above sea level form marine islands, and flat-topped ones form oceanic plateaus.
- Marine Deeps and Trenches – deep, narrow, steep landforms on the ocean bed; the Mariana Trench (Pacific Ocean) is the deepest, at about 11,034 m. (Students should draw and label a cross-sectional diagram of the ocean floor showing these features, similar to Figure 4.2 in the textbook.)

2) Marine deposits found on the ocean floor are mainly of three types:

- Deposits brought from land – pebbles, clay and soil brought by rivers, glaciers, etc. from the continents; deposited mainly on the continental shelf, these are called marine deposits, and layering of these under pressure forms sedimentary rocks.
- Marine oozes – lava and ash from volcanic eruptions along with fine soil particles, mixed with remains of marine plants and animals, forming fine clay-like deposits (about 30% of ocean deposits); useful for understanding marine life and minerals on the ocean bed.
- Anthropogenic (human-induced) deposits – sewage, solid waste, radioactive material, chemicals and plastics disposed of by man; though small in proportion, they are extremely hazardous to marine life and the environment.